

PHARMACEUTICAL IMPORT & DISTRIBUTION MANAGEMENT SYSTEM

PROJECT PROPOSAL

GROUP NO 13

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Background

Pharmaceutical import companies in Sri Lanka act as critical intermediaries between international manufacturers and local healthcare providers (hospitals and pharmacies). Currently, most of these companies rely on manual or semi-digital processes. Orders from clients are received via phone calls, WhatsApp messages, or paper forms, which are then manually entered into spreadsheets. Inventory tracking is often done using basic software or even physical logbooks, making it difficult to track batch-wise stock, expiry dates, and supplier details in real time. The procurement process with foreign suppliers is similarly disconnected, leading to delays, stock inconsistencies, and operational inefficiencies. There is a clear need for an integrated digital system that can automate and streamline the entire supply chain, from client order placement to supplier procurement.

Problem Statement

The main problem is the lack of an integrated computer system. All operations are managed manually or through disconnected files, leading to critical inefficiencies:

1. **Inefficient Order Management:** Orders from hospitals/pharmacies are taken by phone or paper. This leads to wrong entries, duplicate orders, and slow processing.
2. **Poor Inventory Visibility:** The company does not know exactly what medicines are available right now. This causes two problems:
 - Running out of popular medicines (stockouts).
 - Having too much of slow-selling medicines, which can expire and be wasted.
3. **Slow Procurement Process:** The process to order new stock from foreign companies is slow and done with papers. It is not connected to the sales data, so they often order too late.
4. **Lack of Client Self-Service:** Hospitals and pharmacies cannot check stock, place orders online, or see their past orders. They must always call or email, which takes a lot of time for the staff.
5. **Data Silos & Reporting Difficulties:** Sales info, stock details, and supplier data are kept in different Excel files and notebooks. Because of this, making useful reports or smart business decisions is very hard.

Therefore, to address these challenges effectively, there is a critical need to develop a formal, centralized database system integrated with a dedicated online platform.

Objectives

The primary goal of this project is to design and propose a centralized database system integrated with a dedicated online platform that addresses the above problems. The specific objectives are:

1. To develop a computerized system that can manage medicine orders, inventory, and supplier purchases in an efficient and accurate way.
2. To provide real-time information about medicine stock levels, batch expiry dates, and order status to both company staff and client hospitals/pharmacies.
3. To track all sales and procurement data in a centralized database, allowing the management to generate useful reports and make better business decisions.

Scope

A. **Functional Scope**

The system will provide the following key functions ,

1. User Management & Authentication:
 - Allow new Hospitals and Pharmacies to register online as Clients.
 - Allow System Administrators to manage all user accounts.
 - Authenticate users and control access based on their role (Admin, Hospital, Pharmacy).
2. Medicine & Inventory Management:
 - Add, update, and delete medicine details (Medicine Master List).
 - Record batch-wise stock entries, including expiry dates and import costs.
 - Display real-time stock availability to Clients.
 - Generate automatic low-stock alerts for Administrators.
3. Sales Order Processing:
 - Allow Clients to browse the available medicine catalog.
 - Allow Clients to create Sales Orders by selecting items and quantities.
 - Allow Clients to view their order history and track current order status.
 - Automatically update stock levels when an order is approved.
 - Allow Admins to view and update order status (Pending, Approved, Dispatched).
4. Procurement Management:
 - Allow Admins to view low-stock reports.
 - Allow Admins to create Purchase Orders for selected Suppliers.
 - Maintain a complete record of all Purchase Orders and their status.
 - Automatically update stock levels when a Purchase order is Received .
5. Basic Reporting:
 - Generate essential reports: Low-stock reports, Sales Summaries, Expired Batches List, and Pending Orders Report.

B. **Technical Scope**

The system will be a full-stack web application built using

- MySQL for the relational database.
- PHP for the server-side backend logic.
- HTML, CSS, and JavaScript for the frontend user interface.
- The database will be designed using a normalized schema based on an Enhanced Entity-Relationship (EER) Diagram.

C. **Out of Scope**

- The system will not handle online payments or financial transactions.
- Live shipment or delivery tracking features are not included.